

Seyedmahdiar Ziaolhagh

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Summary

An aerospace engineer and graduate researcher who works on instrumentation, mechanical design, vacuum systems, cryogenic testing, vibration analysis, and experimental data processing. Looking for aerospace, mechanical, manufacturing, or R&D engineering roles involving design, testing and analysis.

Education

University of Lethbridge Lethbridge, Alberta, Canada
Master of Science in experimental physics Sept. 2024 – Present

- Thesis: Evaluation of composite material mechanical properties in space and stratospheric environments.

K. N. Toosi University of Technology Tehran, Iran
Bachelor of Engineering in Aerospace Engineering Sept. 2019 – Feb. 2024

- Thesis: Mineral extraction scenarios from an asteroid via optimization considering rotational dynamics.
- GPA: 16.5/20; ranked among the top 10% of aerospace engineering students.

Technical Skills

CAD and Design:	SolidWorks, mechanical component design, 3D printing, laser cutting, sheet-metal
Simulation and Analysis:	Abaqus, ANSYS STK, structural analysis, thermal analysis, vibration analysis, finite element analysis (FEA)
Programming:	Python, MATLAB, NumPy, SciPy, Pandas, data processing, signal processing, FFT analysis
Laboratory and Testing:	Vacuum systems, residual gas analyzer, cryogenic testing, composite material testing, interferometry, mechanical testing
Electronics and Tools:	LTspice, RGASoft, instrumentation, experimental setup design

Professional Experience

Research Assistant Jan. 2025 – Present
University of Lethbridge Lethbridge, Alberta, Canada

- Design and validation of a custom **three stage** three stage vacuum chamber for mechanical Characterization of CFRP at cryogenic and room temperatures.
- Designed experimental setups and testing methodologies to characterize composite materials for aerospace applications in simulated space and stratospheric environments.
- Characterization of the CFRP outgassing behaviour using a Residual Gas Analyzer (RGA).
- Developed signal processing algorithms and analyzed **300 and more** vibration experiments to estimate mechanical properties and quantify experimental uncertainty.
- Designed a 3-DOF system for surface scanning via LASER interferometry to find surface roughness.

Teaching Assistant Jan. 2025 – May 2025
University of Lethbridge Lethbridge, Alberta, Canada

- Delivered laboratory instruction for undergraduate Engineering Physics courses covering experimental methods, uncertainty analysis, and scientific reporting.
- Mentored students during laboratory sessions, helping them troubleshoot experiments, interpret results, and improve technical reports.
- Evaluated laboratory performance and provided constructive technical feedback to improve students' analytical and experimental skills.

Mechanical Component Designer Apr. 2023 – Nov. 2024
Fara-Damay-E-Alborz Ltd. Tehran, Iran

- Designed custom mechanical components and assemblies according to customer requirements, manufacturing constraints, and functional specifications.
- Produced detailed CAD models and engineering drawings to support fabrication and manufacturing.
- Collaborated directly with customers to optimize component performance, manufacturability, and production cost.

Teaching Assistant

K. N. Toosi University of Technology

Sept. 2023 – Dec. 2023

Tehran, Iran

- Led weekly tutorial sessions for undergraduate Dynamics courses, strengthening students' understanding of engineering mechanics.
- Designed assignments and examinations, graded coursework, and provided academic mentoring throughout the semester.
- Guided students in solving real-world engineering mechanics problems using analytical methods.

Selected Engineering Projects

Vacuum and Cryogenic Composite Testing System

- Designed and built a laboratory system for mechanical testing of composite materials under cryogenic, vacuum, and room-temperature conditions.
- Integrated thermal, mechanical, and vacuum subsystems to simulate environmental conditions encountered during aerospace and space missions.
- Performed structural design using SolidWorks, finite-element analysis using Abaqus, and automated data processing using Python.

3-degree-of-freedom (DOF) System for Interferometric Surface Profiling

- Designed a fully automated **3-DOF robotic platform** for interferometric measurement of engineering surfaces.
- Developed positioning algorithms to automate surface scanning and improve measurement repeatability for precision metrology.
- Implemented Python-based data processing to reconstruct high-resolution surface profiles from interferometric measurements.

Heat Sink Design for Electrical Harness

- Designed a thermal management solution for an electrical harness using SolidWorks and LTspice.
- Evaluated heat dissipation performance and optimized the design for improved thermal reliability.

Publications and Conference Presentations

Journal Publication

- **Mineral Extraction Scenarios from an Asteroid via Optimization Considering Rotational Dynamics**
IEEE Access, 2025.

Authors: Seyedmahdiar Ziaolhagh and Mahdi Jafari-Nadoushan.

[DOI](#)

Conference Presentations

- **Characterization Of Composite Materials For Space Environments**
Canadian Astronomical Society Annual Meeting (CASCA 2026), Montreal, QC, Canada. [Poster PDF](#)
- **Technology Development for Space-Based Far-Infrared Spectroscopy and Interferometry**
Canadian Astronomical Society Annual Meeting (CASCA 2026), Montreal, QC, Canada. [Poster PDF](#)
- **When Your Equipment Lies to You and What to Do About It: How to Improve Experimental**
University of Lethbridge Summer Research Showcase, Lethbridge, AB, Canada, 2025. [Poster PDF](#)
- **Laser Metrology for Material Characterization Experiments**
University of Lethbridge Summer Research Showcase, Lethbridge, AB, Canada, 2025. [Poster PDF](#)

Certificates

- NASA Open Science 101 Curriculum, NASA, June 2025. [View certificate](#)
- NSERC SMART CREATE Entrepreneurship Workshop, University of Alberta, March 2025. [View certificate](#)
- Supervised Machine Learning: Regression and Classification, DeepLearning.AI and Stanford University, Coursera, Jul. 2023. [View certificate](#)
- Commercial Pilot Ground Course, Homa Aviation Center, Tehran, Iran, March. 2022. [View certificate](#)
- Certificate of participate in the world robotic Olympiad, Greater Noida, India Nov. 2016. [View certificate](#)

Honours and Awards

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- Ranked among the top 10% of aerospace engineering students during B.Sc. program, 2019–2024.
 - Third place, World Robot Olympiad qualifiers in Iran, Nov. 2016.

Volunteer and Outreach Experience

STEM Outreach Presenter, University of Lethbridge Apr. 2026

- Delivered a presentation on physics applications to encourage high school students to pursue STEM fields.
- Designed and demonstrated hands-on physics experiments for student attendees.

Laboratory Tour Guide, University of Lethbridge Nov. 2025

- Introduced high school students to instrumentation laboratories and explained experimental equipment and research activities.

SolidWorks Instructor, University of Lethbridge May 2025 – Sept. 2025

- Designed and delivered SolidWorks training from beginner to advanced levels for summer students.

Scientific Researcher and Technical Writer, OWJ(A monthly, Aerospace-Based student magazine) , Sharif University of Technology, Tehran, Iran Sep 2022 – Jan. 2025

- Conducting research, interviewing experts in the field, and writing informative articles that engage and educate readers about the latest advancements in space exploration and technology.

Languages

Persian: Native

English: Fluent